

The Growing Capacity and Latency Cost of CPU Caching

Hong Chan Lee Justin Kuethe

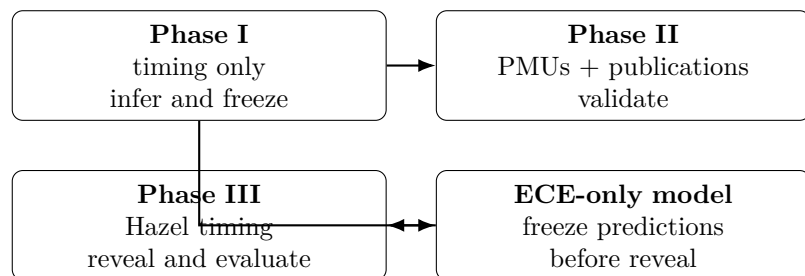
NC State ECE — Fall 2026

<https://github.com/NovaCorz/CSC591-Project1>

Question, machine timeline, and blind sequence

Experimental question

Can timing-only measurements recover cache behavior on heterogeneous systems, predict later generations, and survive PMU, published-data, and held-out Hazel checks?



Primary contributor: Hong Chan Lee (phase logic).

- 10^6 timed samples at every full tested point.
- Failed/noisy attempts retained; unsupported fields remain explicit.

Chronology used in plots

Year	Machine / target	Generation	ISA
2014	Sunbird; Hazel	Haswell	x86
2015	Charnwood	Skylake	x86
2017	Ookay; Upgrade	Kaby; Coffee Lake	x86
2018	Crux	Coffee Lake Refresh	x86
2019	Skylark; Hazel	Zen 2; Cascade Lake	x86
2020	Thunderbird	Neoverse N1	Arm
2021	Hazel	Ice Lake-SP	x86
2022	Hazel	Genoa / Zen 4	x86
2023	Artemisia	Sapphire Rapids	x86
2024	Hazel	Turin / Zen 5	x86

Primary contributor: Hong Chan Lee (chronological integration).

Finding. Coverage: eight ECE machines establish the timing-only training set; five held-out Hazel generations test transfer after Phase-II validation and the ECE-only prediction freeze.

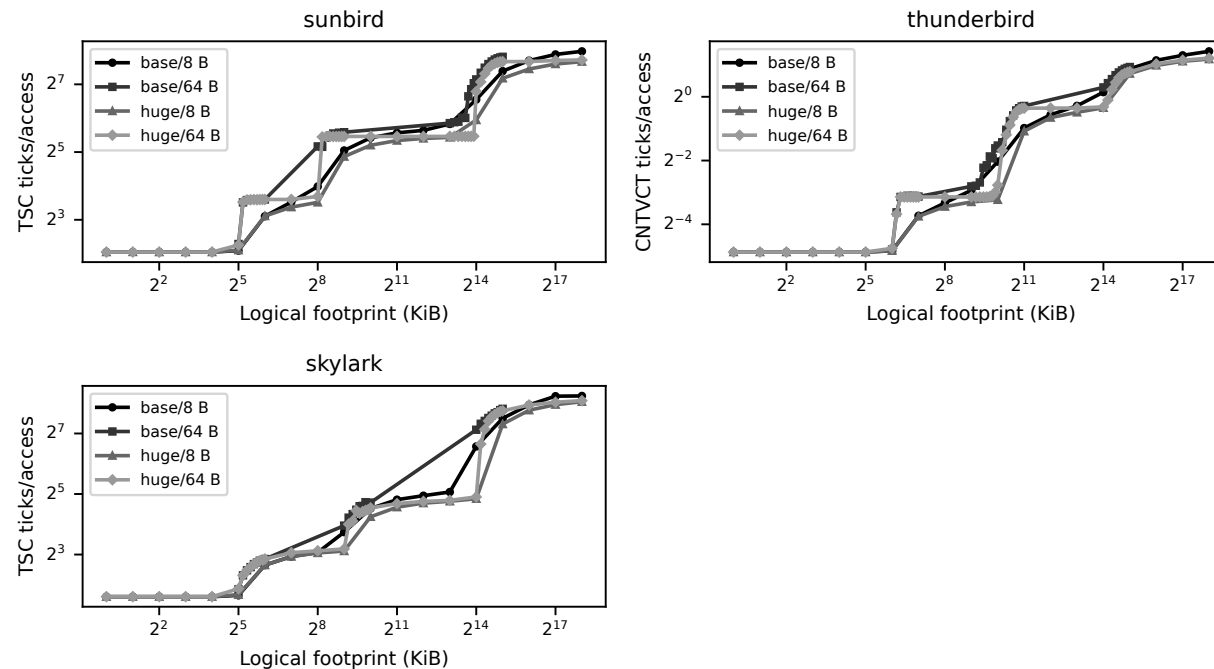
Reverse engineering 1: hierarchy and capacity

Random permutation of W bytes
 $A \rightarrow C \rightarrow F \rightarrow \dots$

Pin one core; warm; sweep W
1 KiB–256 MiB; dense near edges

Time dependent loads
plot distributions and median latency/ W

Plateaus $\Rightarrow$ timing classes
repeatable steps $\Rightarrow$ capacity brackets



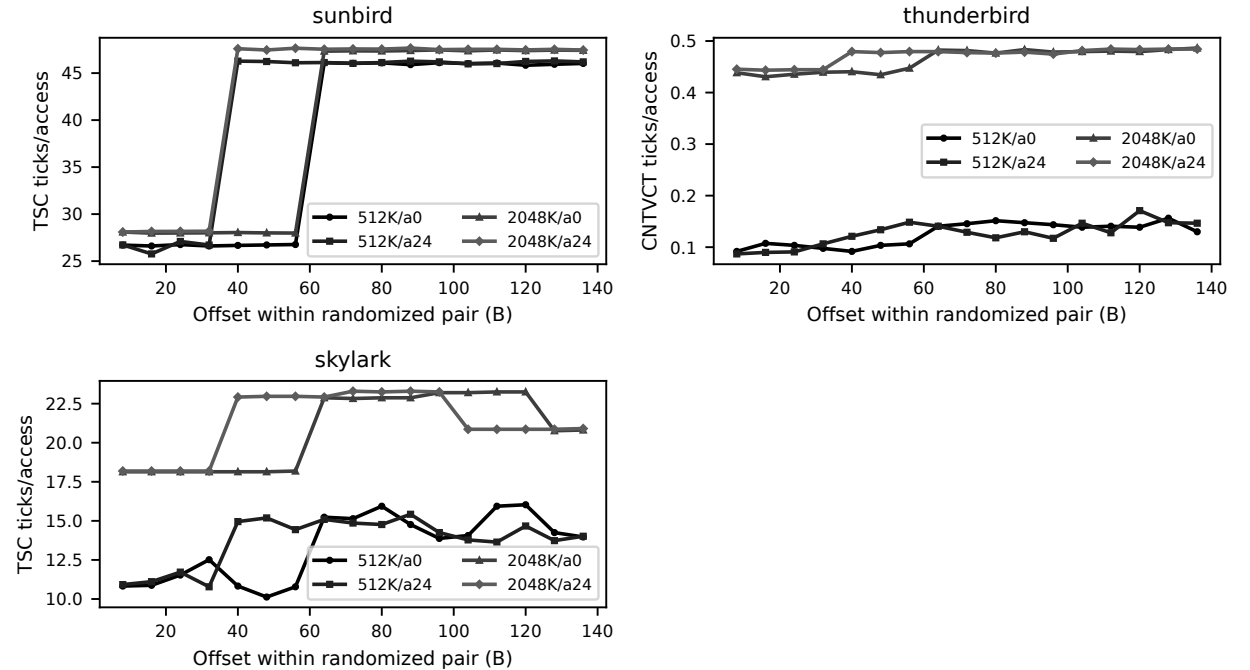
Evidence. Representative Sunbird, Thunderbird, and Skylark capacity sweeps; the final table covers all machines.

Primary contributor: Hong Chan Lee (diagram and evidence).

- 1) **Vary:** logical footprint W
- 2) **Hold/control:** core, timer, page policy, spacing, seed cohort, dependent read-only loop
- 3) **Measure:** ticks/access and P5, median, IQR, P95 for 10^6 samples
- 4) **Seek:** repeatable plateaus and adjacent distribution-separated increases
- 5) **Infer:** effective hierarchy classes and operational capacity bounds
- 6) **Validate later:** regular traversal/prefetch controls, repeats, then Phase-II PMUs and published values

Finding. All machines expose multiple timing classes. Broad deeper transitions are retained as brackets and do not alone prove a physical cache level.

Reverse engineering 2: spatial granularity / line size



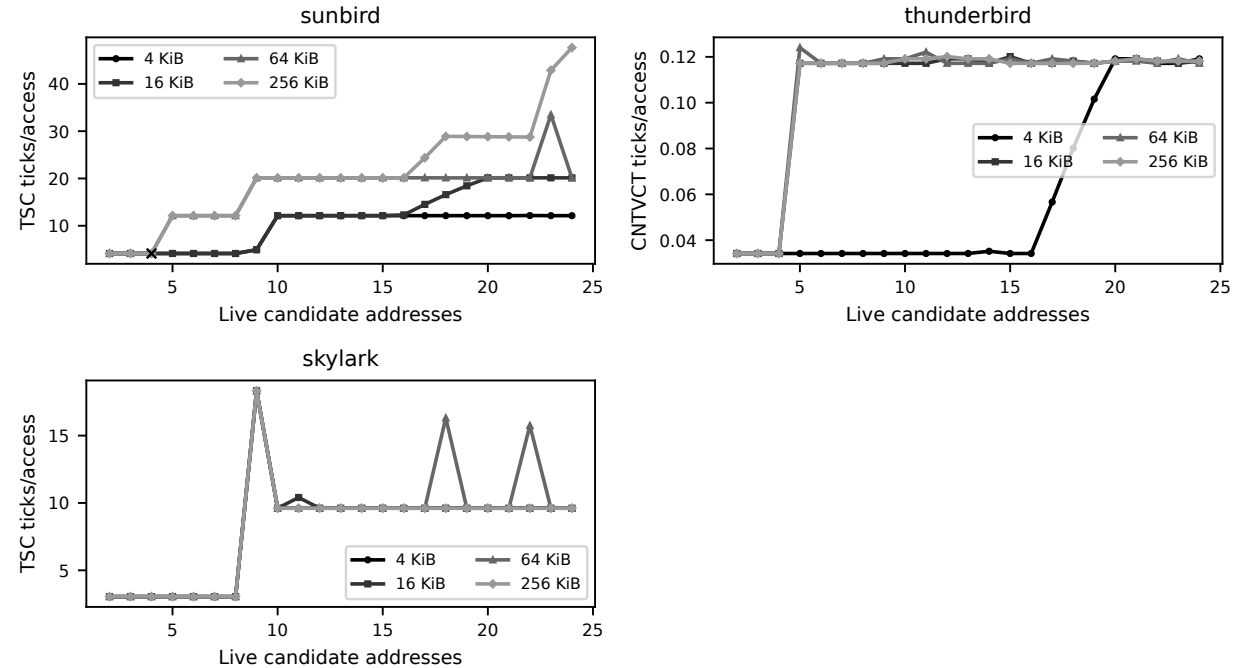
- 1) **Vary:** offset δ and alignment α
- 2) **Hold/control:** footprint, core, timer, warm-up, random pair order, sample count
- 3) **Measure:** latency distributions for every candidate offset
- 4) **Seek:** the first repeated IQR-separated change below/at/above the boundary
- 5) **Infer:** smallest timing-visible spatial granularity
- 6) **Validate later:** new seeds/scales, regular/random controls, Phase-II stride PMUs and published line size

Evidence. Representative offset sweeps for Intel, Arm, and AMD ECE systems.

Primary contributor: Hong Chan Lee (diagram and evidence).

Finding. All eight ECE machines and all five Hazel generations reproduce a 64 B boundary; separate physical line sizes at every level remain isolated.

Reverse engineering 3: associativity



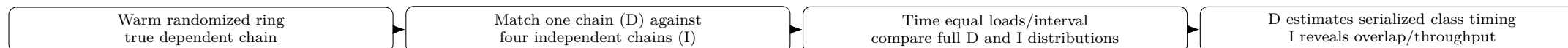
- 1) **Vary:** number of live candidate-conflicting lines
- 2) **Hold/control:** core, footprint class, candidate stride, alignment cohort, dependent traversal
- 3) **Measure:** ticks/access and eviction-sensitive distribution for 10^6 intervals
- 4) **Seek:** sharp reproducible increase at w or $w + 1$ across controls
- 5) **Infer:** conditional effective associativity; cross-check $S = C/(AB)$
- 6) **Validate later:** repeat periods/alignments and compare Phase-II L1D-miss thresholds and publications

Evidence. Representative conflict-count sweeps; physical congruence is not assumed.

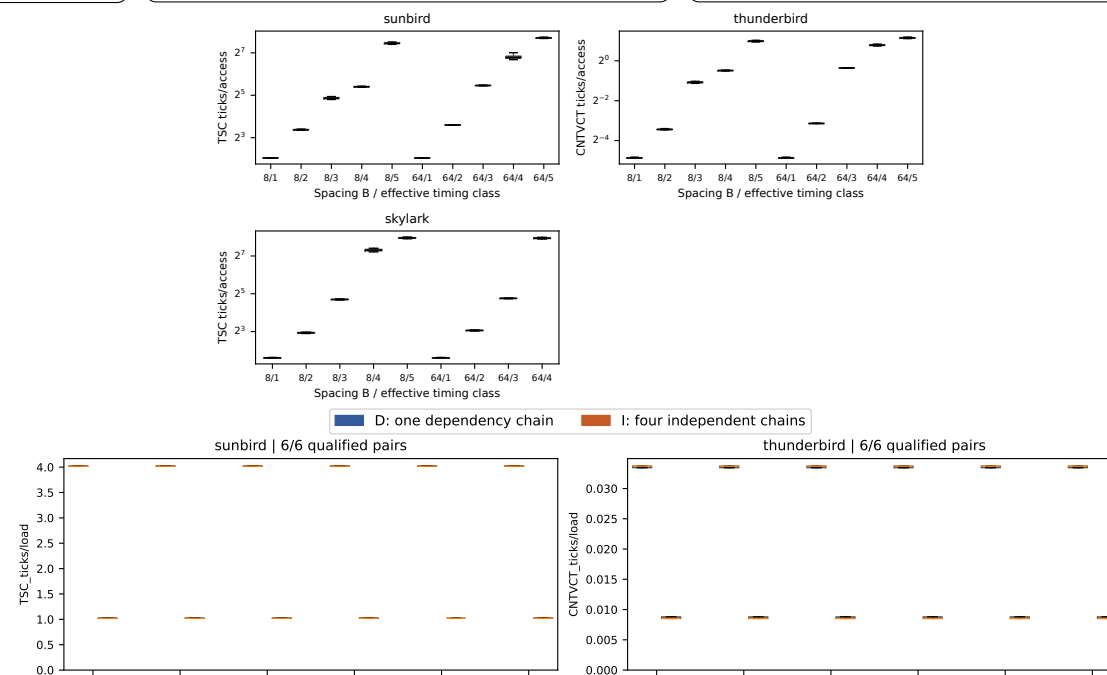
Primary contributor: Hong Chan Lee (diagram and evidence).

Finding. ECE L1-like candidates are 8 ways except Thunderbird (4) and Artemisia (12). Only Crux and Thunderbird retain conditional deeper-way candidates.

Reverse engineering 4: hit latency and throughput diagnostic



- 1) **Vary:** resident timing class; D versus I dependency structure
- 2) **Hold/control:** same ring, core, timer, warm-up, four loads/iteration and 4096 loads/interval
- 3) **Measure:** native ticks/load distributions and same-core representative medians
- 4) **Seek:** stable warmed class; consistently lower I time indicates overlap
- 5) **Infer:** L1-like dependent class latency; never substitute I throughput as latency
- 6) **Validate later:** timer/loop controls, six D/I pairs per host, publications where comparable



Evidence. Representative timing classes (top) and matched dependency diagnostic (bottom).

Primary contributor: Hong Chan Lee (diagram and evidence).

Finding. All 48 ECE and all 30 Hazel D/I pairs qualify; independent chains are faster in every pair, confirming that low throughput time is not individual-load latency.

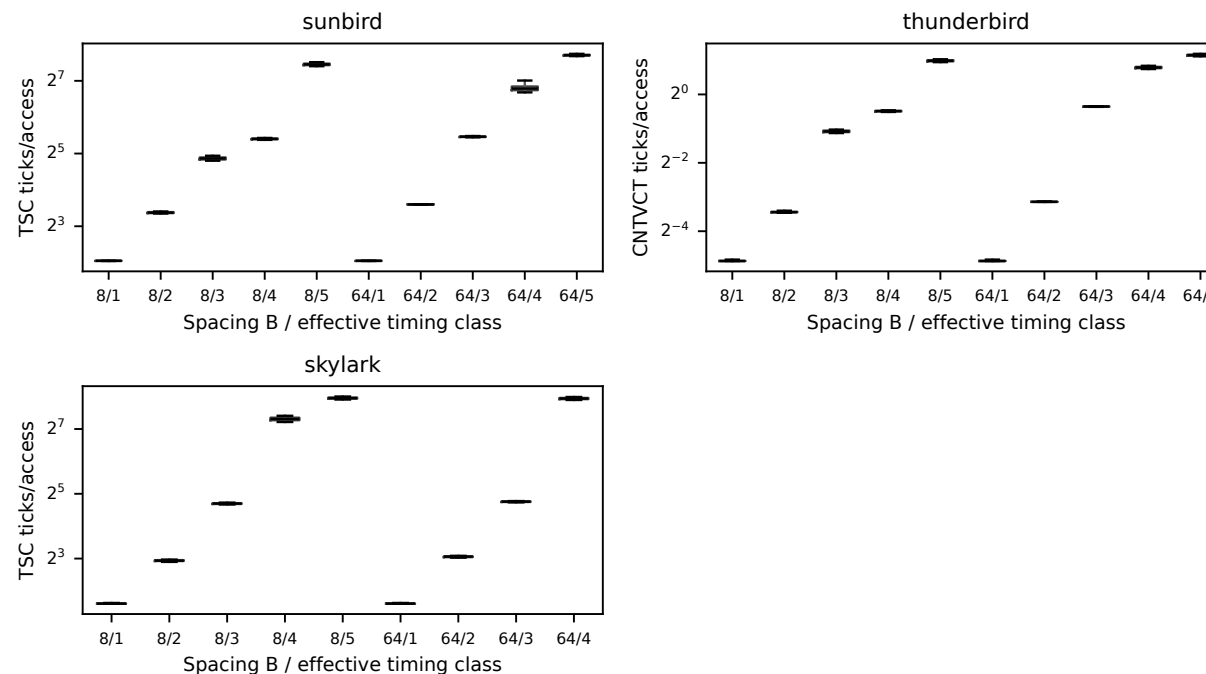
Reverse engineering 5: miss / next-level latency

Choose a warmed timing class

Increase footprint or apply controlled pressure
while retaining the same core/timer

Measure dependent class
and first-reload distributions

Separated next class – prior class
⇒ observable increment only



- 1) **Vary:** working-set class and eviction pressure
- 2) **Hold/control:** CPU, timer, dependency chain, spacing, page policy, warm-up
- 3) **Measure:** full distributions and same-CPU median increments between classes
- 4) **Seek:** repeatable separated next timing mode after the capacity departure
- 5) **Infer:** effective next-class increment; physical hit/miss label only with proven residency
- 6) **Validate later:** capacity correlation, Phase-II PMU boundaries, published data and residency controls

Evidence. Same-CPU class representatives show observable increments and overlapping/noisy deeper modes.

Primary contributor: Hong Chan Lee (diagram and evidence).

Finding. The measurements establish timing-class increments, but do not isolate physical L1 miss, L2/LLC hit, or LLC-to-memory penalties; those numerical series remain unsupported.

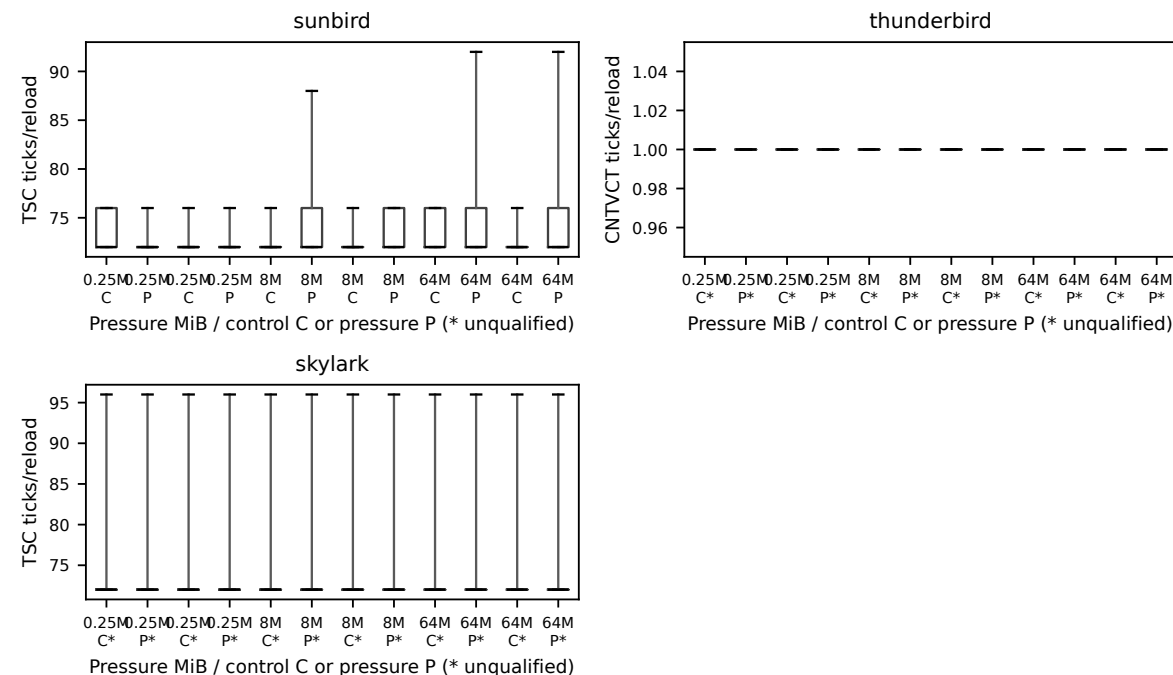
Reverse engineering 6: inclusion / exclusion behavior

Warm target on pinned core

Helper core traverses pressure region
no-load handshake is the control

Reload target; compare paired distributions

Only proven lower-only eviction
+ upper survival/removal
supports a policy inference



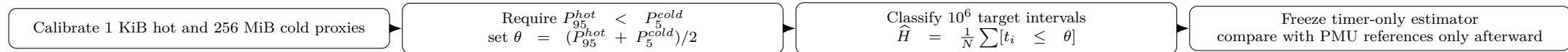
- 1) **Vary:** helper pressure size and target/pressure pairing
- 2) **Hold/control:** target core, helper socket placement, handshake, timer, target address, warm-up
- 3) **Measure:** target reload distributions with pressure versus no-load control
- 4) **Seek:** repeatable reload shift after demonstrated lower-only eviction
- 5) **Infer:** inclusive evidence if upper copy disappears; survival may support non-inclusion only with residency proof
- 6) **Validate later:** multiple targets/controls plus post-freeze PMU and published organization

Evidence. Representative paired pressure/control reload distributions.

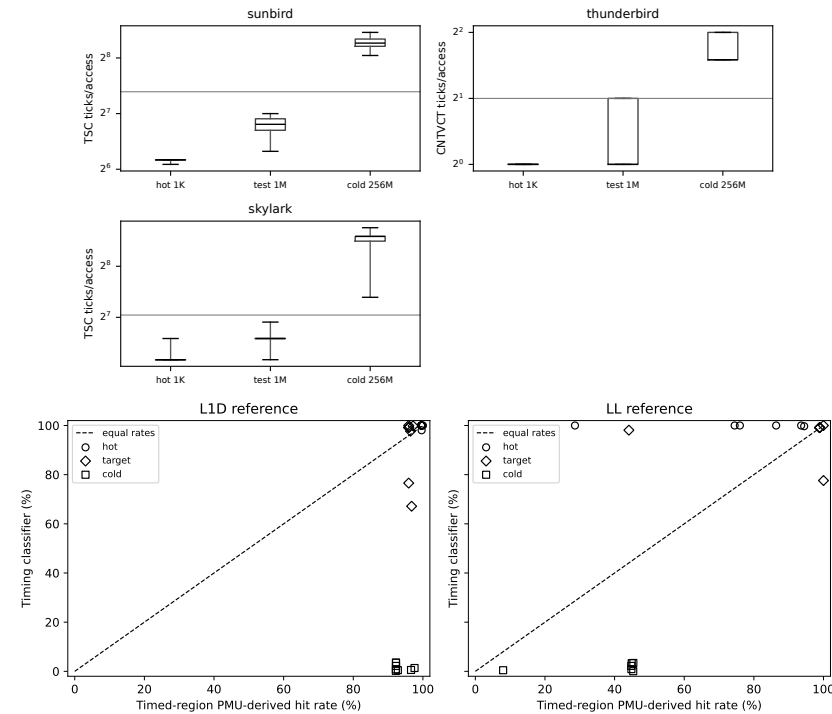
Primary contributor: Hong Chan Lee (diagram and evidence).

Finding. Qualified comparisons are uneven and do not prove lower-only eviction or upper residency. Inclusion/exclusion therefore remains uncertain on every machine.

Reverse engineering 7: software-only residency metric



- 1) **Vary:** footprint: hot, 1 MiB target, cold
- 2) **Hold/control:** logical CPU, access pattern, seed, timer, batch size, warm-up
- 3) **Measure:** complete timing distributions and fraction below frozen threshold
- 4) **Seek:** separated calibration distributions and stable threshold sensitivity
- 5) **Infer:** software residency proxy, or unresolved when calibration overlaps
- 6) **Validate later:** Phase-II PMU-derived reference rates; preserve disagreement and event limitations

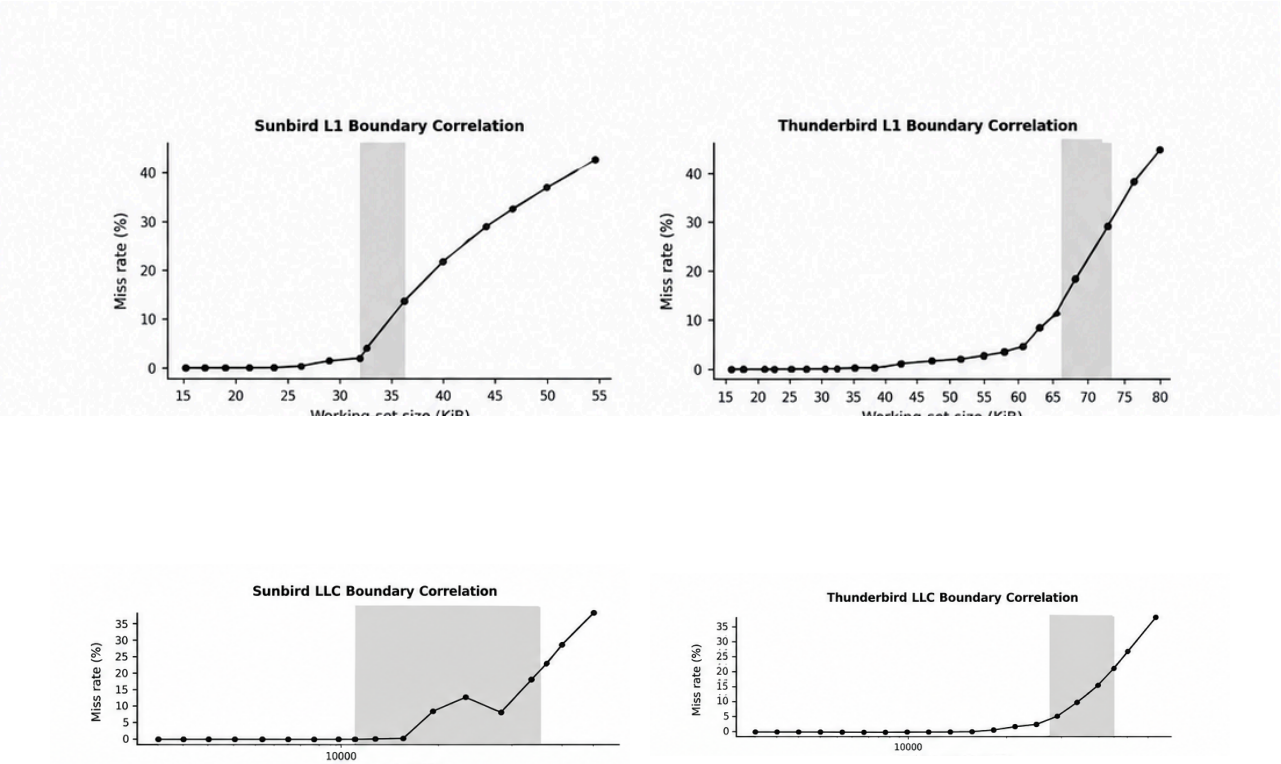


Evidence. Calibration/classes (top) and timing-estimator versus PMU references (bottom).

Primary contributor: Hong Chan Lee (diagram and estimator evidence); Justin Kueth (Phase-II PMU inputs).

Finding. The estimator is portable but threshold-sensitive; its percentages are timing classifications, not hardware hit probabilities.

Phase II: PMU validation after the timing freeze



Evidence. Representative L1 and LLC correlations compare the timing edge with normalized cache-event changes.
Primary contributor: Justin Kuethe.

Target	PMU evidence	Outcome
Capacity	L1/LL transitions	L1 clean 7/8; LLC clean 4/6 available
Line/stride	L1D misses/access at 56/64/72 B	Broad responses; Sunbird unavailable
L1 associativity	miss/access $w - 1, w, w + 1$	at Edge at all eight conditional candidates
L2 associativity	generic LL events	Not a level-specific validation

Primary contributor: Justin Kuethe.

- Workloads and 10^6 repetitions match Phase I.
- Counters were batched where simultaneous collection would multiplex.
- Architecture-specific mappings and unavailable events remain explicit.

Finding. PMUs corroborate several L1 timing boundaries and conflict candidates, but architecture-specific event mappings and unavailable events limit deeper-level claims.

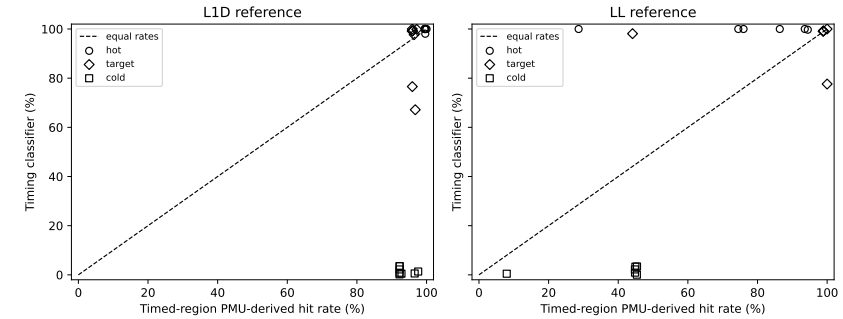
Phase II: all eight counters across three workloads

High → low, normalized per dependent access. Ar=Artemisia, Ch=Charnwood, Cr=Crux, Oo=Oakay, Sb=Sunbird, Sk=Skylark, Th=Thunderbird, Up=Upgrade.

Event	L1-resident	LLC-sized	Larger than LLC
cycles	Ch>Cr>Up>Oo>Sk>Ar>Sb>Th	Cr>Oo>Sk>Up>Ch>Th>Ar>Sb	Sk>Cr>Up>Oo>Th>Sb>Ar>Ch
instructions	Cr>Up>Oo>Sk>Ch>Ar>Sb>Th	Th>Up>Sb>Sk>Oo>Cr>Ch>Ar	Th>Up>Sk>Oo>Cr>Sb>Ch>Ar
L1D-read-access	Up>Cr>Sk>Oo>Sb>Ch>Ar>Th	Sk>Th>Sb>Ar>Up>Cr>Ch>Oo	Sk>Th>Sb>Cr>Up>Ar>Ch>Oo
L1D-read-miss	Ch>Sk>Cr>Oo>Up>Sb>Th>Ar	Ch>Sb>Oo>Up>Cr>Ar>Sk>Th	Ch>Up>Cr>Oo>Sb>Ar>Sk>Th
LL-read-access	Ch>Oo>Up>Cr>Sb>Ar	Ch>Cr>Up>Oo>Sb>Ar	Ch>Cr>Oo>Up>Sb>Ar
LL-read-miss	Oo>Ch>Cr>Up>Ar>Sb	Ch>Oo>Cr>Up>Sb>Ar	Ch>Oo>Cr>Up>Sb>Ar
DTLB-read-miss	Ch>Th>Sk>Sb>Up>Cr>Oo>Ar	Th>Ch>Sb>Up>Oo>Cr>Ar>Sk	Th>Ch>Oo>Up>Cr>Sb>Sk>Ar
cache-misses	Oo>Ch>Th>Up>Cr>Sk>Sb>Ar	Ch>Oo>Cr>Up>Sk>Th>Sb>Ar	Ch>Oo>Cr>Up>Sk>Th>Sb>Ar

Primary contributor: Justin Kuethe.

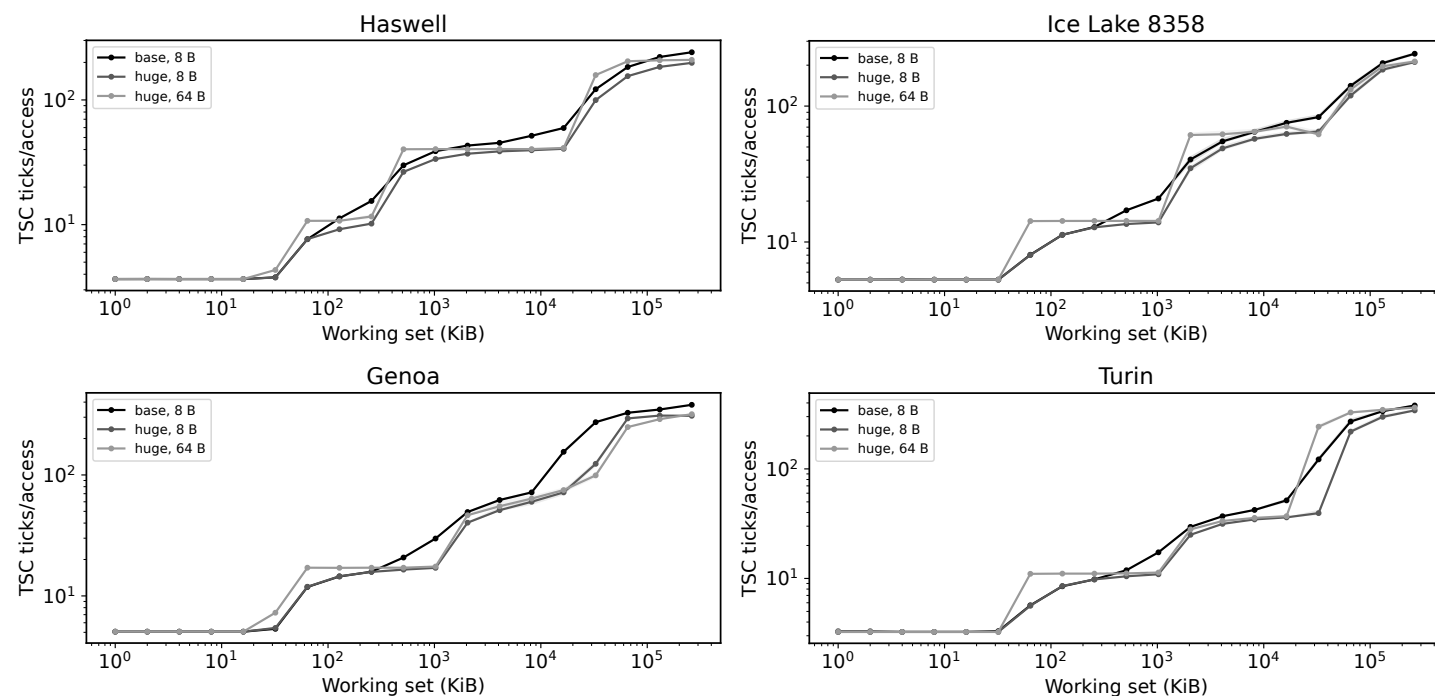
- Exact exposed events: cycles, instructions, L1D-read-access, L1D-read-miss, LL-read-access, LL-read-miss, DTLB-read-miss, cache-misses.
- LL ranks contain six machines because generic LL events are unavailable on Skylark and Thunderbird.
- Generic mappings differ by architecture; the rankings describe exposed event populations.



Evidence. Software estimator versus two matched PMU reference definitions.

Primary contributor: Hong Chan Lee (comparison); Justin Kuethe (PMU acquisition).

Phase III: Hazel replication on five generations



Evidence. Readable capacity evidence for Haswell, Ice Lake-SP, Genoa, and Turin; Cascade Lake's unresolved deeper result remains in the table.

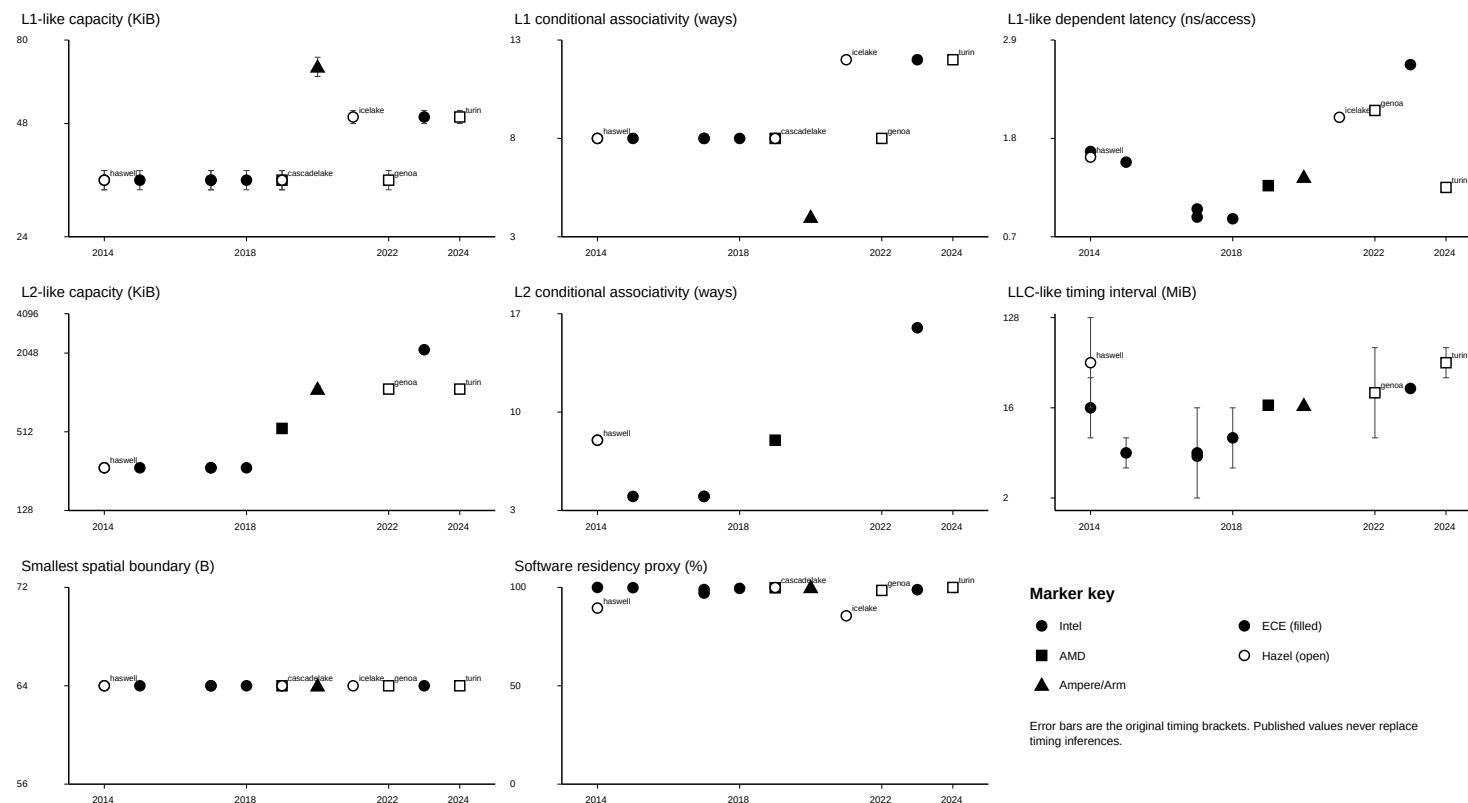
Primary contributor: Hong Chan Lee.

Generation	L1 KiB	w	L2 KiB	LLC MiB
Haswell	32–36	8	256–288	16–128
Cascade	32–36	8	–	–
Ice Lake	48–52	12	–	–
Genoa	32–36	8	1024–1152	8–64
Turin	48–52	12	1024–1152	32–64

Primary contributor: Hong Chan Lee.

- Same native Phase-I methods and 10^6 samples/full point.
- Slurm job, CPU model, binding, topology, compiler/flags, seed, timestamps and hashes retained.
- Five generations span 2014–2024 and include AMD.
- All show 64 B; all 30 D/I pairs qualify; inclusion remains uncertain.

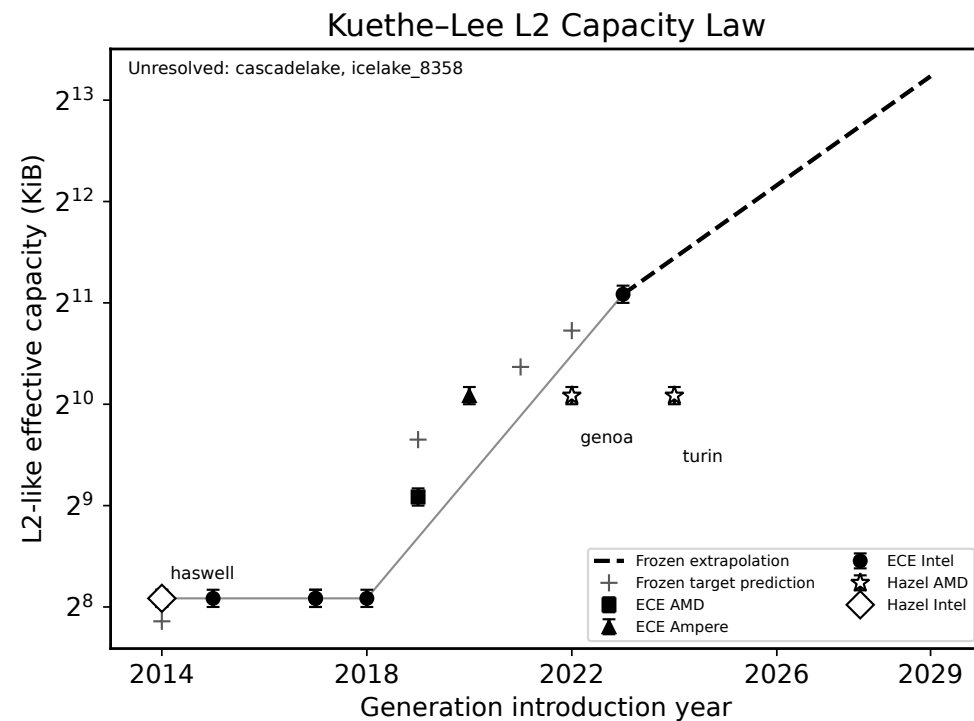
Cross-generation cache evolution and support limits



Evidence. ECE observations (filled) and Hazel observations (open), ordered by processor-generation introduction year; capacity error bars preserve timing brackets.

Primary contributor: Hong Chan Lee. **Explicitly unsupported physical series:** L1 miss penalty; L2 hit/miss penalty; LLC associativity, hit latency and memory penalty; exact sharing scope. Inclusion/exclusion is uncertain for every system. These fields are not replaced with specifications.

Frozen prediction versus Hazel reveal



Evidence. Lab observations are filled; the frozen ECE-only model is dashed; held-out Hazel values use open markers.

Primary contributor: Hong Chan Lee.

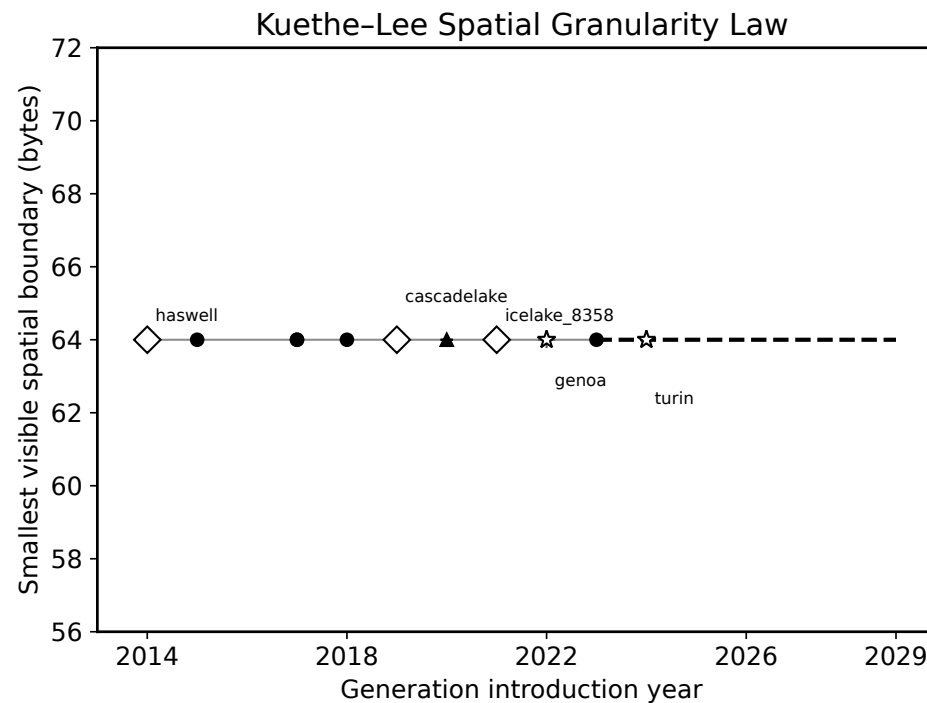
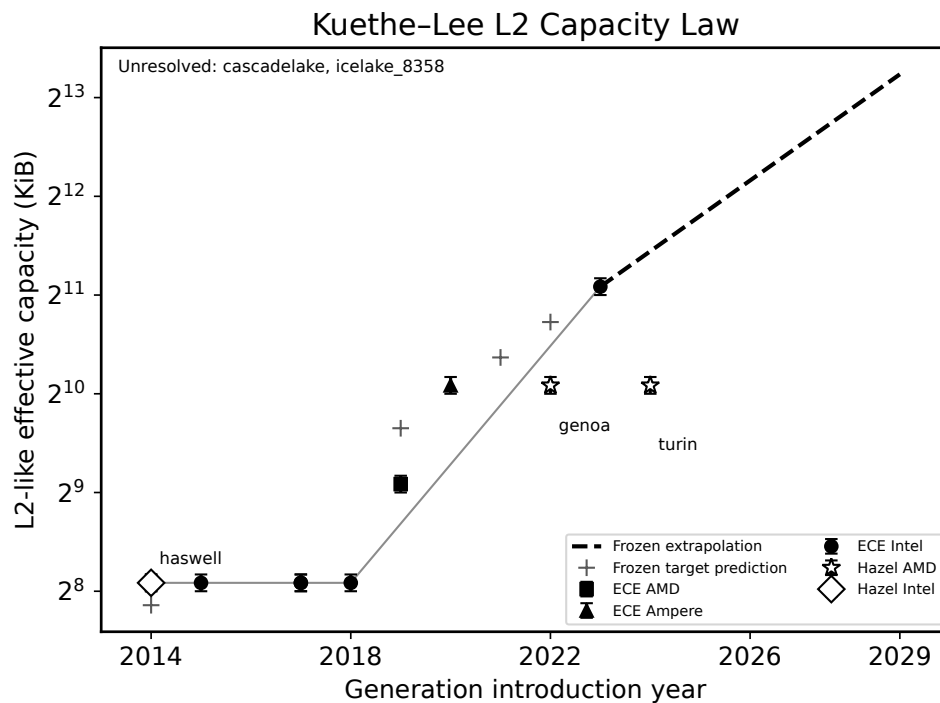
Target	Prediction KiB	Hazel KiB	Assessment
Haswell	232.1	271.5	range overlaps
Cascade	804.0	unresolved	not scored
Ice Lake	1321.5	unresolved	not scored
Genoa	1694.3	1086.1	36.0% high
Turin	2785.0	1086.1	156.4% high

Primary contributor: Hong Chan Lee.

- Haswell and Genoa overlap their broad heuristic model ranges.
- Turin's measured bracket is disjoint from the saved range and falsifies the future-year transfer test.
- Predictions were not refit after Hazel reveal.

Finding. The constant spatial prediction succeeds on all five generations; the four-ECE-server L2 growth model transfers poorly to newer AMD.

Our laws, five-year prediction, and scaling limits



Evidence. Both named laws retain the original ECE-only dashed predictions and add Hazel as held-out markers.

Primary contributor: Hong Chan Lee.

Kuethe-Lee L2 Capacity Law

$$\log_2 C_{\text{KiB}} = \log_2(2172.232) + 0.35849(y - 2023).$$

ECE servers imply 2.79-year doubling; Turin weakens/falsifies transfer. SRAM density, access energy, wire delay, chiplets, sharing and SKU goals favor flattening or piecewise growth.

Kuethe-Lee Spatial Granularity Law

$B(y) = 64$ bytes. All eight ECE and all five Hazel observations support this scoped behavior law. Coherence, sectors or bandwidth tradeoffs could cause a future design step.

2029 prediction: L2-like 9.421 MiB [3.737, 23.752]; spatial boundary 64 B.

Conclusions

- Timing exposes effective boundaries and classes, not every physical field.
- PMUs support L1 capacity/conflict behavior but vary by event mapping.
- The 64 B law transfers; L2 growth does not transfer cleanly.

Material limitations

- Deeper hit/miss penalties and LLC ways remain unisolated.
- Inclusion and sharing remain unresolved.
- Broad brackets mix cache, translation, replacement and memory effects.

Contributions

- **Hong Chan Lee:** Phase I, Phase III, estimator and competition code/runs/analysis.
- **Justin Kuethe:** Phase II code/runs/analysis; report template, repository, Overleaf and slide infrastructure.

Reproducibility: commands, seeds, compilers/flags, affinity/idle evidence, Slurm metadata, raw distributions, failed/noisy attempts and hashes are retained and traceable.

Competition freeze: tag competition-v1; commit 62d3d2b1b8539303772e2783c575cd0972df3335. Scorecard: 70 candidates (56 Yes*, 11 Hold, 3 No).

<https://github.com/NovaCorz/CSC591-Project1>

Primary references

- NC State ECE, Project 1, Sections 8.1–8.6 and 9–15.
- Intel, Intel 64 and IA-32 Architectures Optimization Reference Manual, Table 2-14.
- A. Fog, The Microarchitecture of Intel, AMD, and VIA CPUs, May 2026.
- Intel processor specification and Xeon Scalable generation pages cited per system in the report.
- AMD EPYC 7002 tuning guide, EPYC 9004 tuning guide, and 5th Gen EPYC architecture guide.
- Ampere Computing, Altra 64-Bit Multi-Core Processor Datasheet, Sections 2.3–2.5.

Questions?